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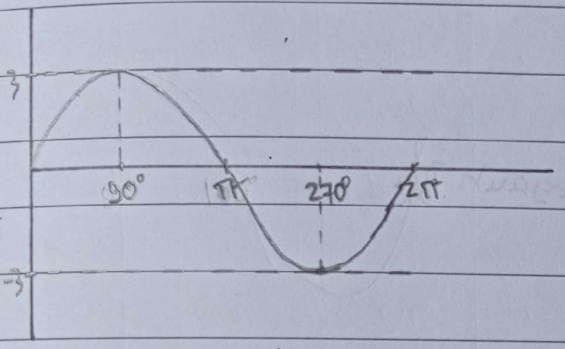
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Latihan 2

A. Tentukan Amplitudo, periode dan gambarkan grafik fungsi trigonometri

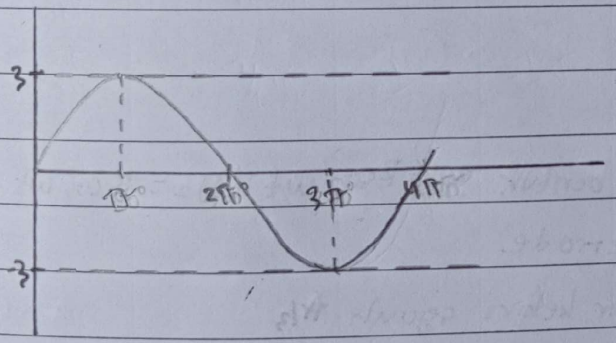
1) $y = 3 \sin x$

- Amplitudonya : 3 satuan
- Periode : $\frac{360^\circ}{1} = 360^\circ = 2\pi$



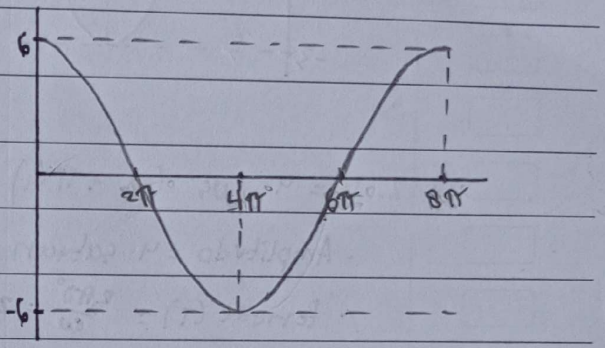
2. $y = 3 \sin 0{,}5x$

- Amplitudo : 3 satuan
- Periode $= \frac{360^\circ}{\frac{1}{2}} = 720^\circ = 4\pi$



3. $y = 6 \cos \frac{x}{4}$

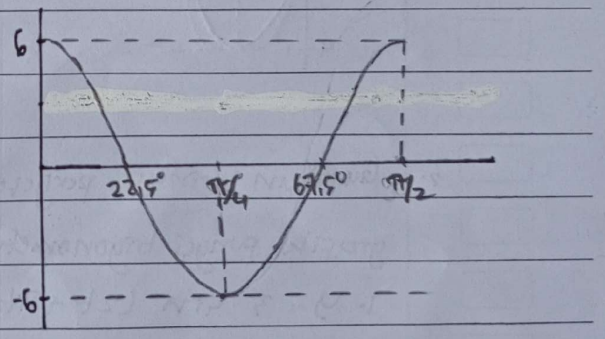
- Amplitudo = 6 satuan
- Periode $= \frac{360}{\frac{1}{4}} = 1440^\circ = 8\pi$



4. $y = 6 \cos 4x$

Amplitudo = 6 satuan

Periode $= \frac{360}{4} = 90 = \pi/2$



B. Gunakan prinsip pergeseran dari bentuk $y_1 = R \sin x$ atau $y_2 = R \cos x$ untuk grafik fungsi trigonometri

1. $y = 3 \sin (x + \pi/3)$ → bergeser kekiri sejauh $\pi/3$

- Amplitudo = 3 satuan
- Periode (T) $= \frac{360^\circ}{1} = 360^\circ = 2\pi$

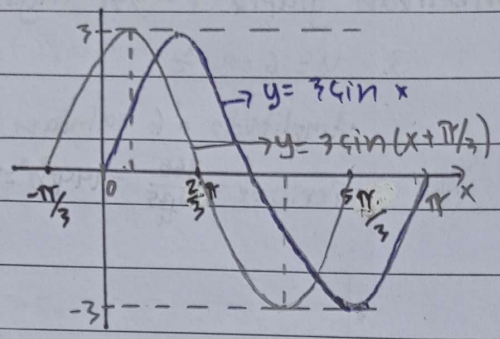
→ $\pi - \pi/3 \Rightarrow 2\pi - \pi/3 = 5\pi/3$

- dari $-\pi/3$ sampai $5\pi/3$

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• Grafik $y = 3 \sin (x + \pi/3)$

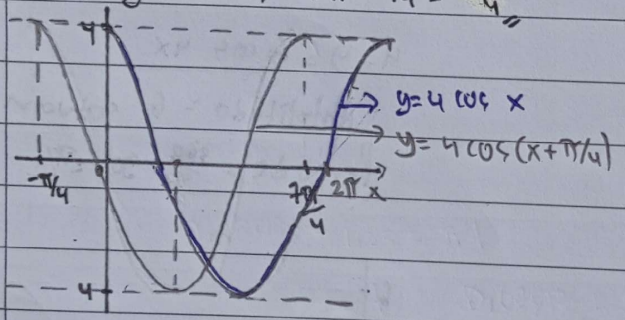


2. $y = 4 \cos (x + \pi/4)$ → kekiri sejauh $\pi/4$

• Amplitudo = 4 satuan

• Periode (T) = $\frac{360^\circ}{1} = 360^\circ = 2\pi$

• Pergeseran ⇒ $2\pi - \pi/4 = \frac{7\pi}{4}$



3. Gunakan prinsip pergeseran dari bentuk $y_1 = R \sin \omega t$ / $y_2 = R \cos \omega t$ untuk grafik fungsi trigonometri satu periode.

1. $y = 3 \sin (2t + \pi/3)$ → geser kekiri sejauh

• Amplitudo 3 satuan

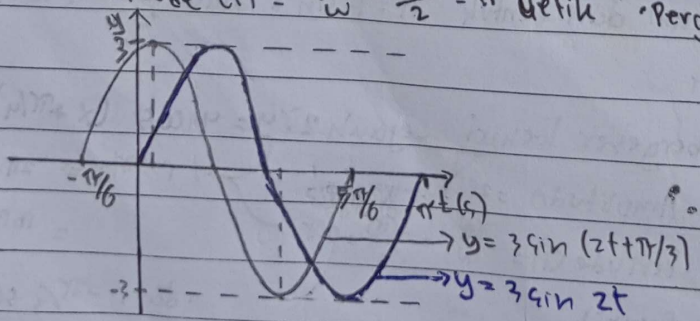
• kecepatan sudut $\omega = 2$

• Periode (T) = $\frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi$ detik

• Pergeseran kekiri sejauh

$2t + \pi/3 = 0$

$t = -\pi/6$ sampai $5/6 \cdot \pi$



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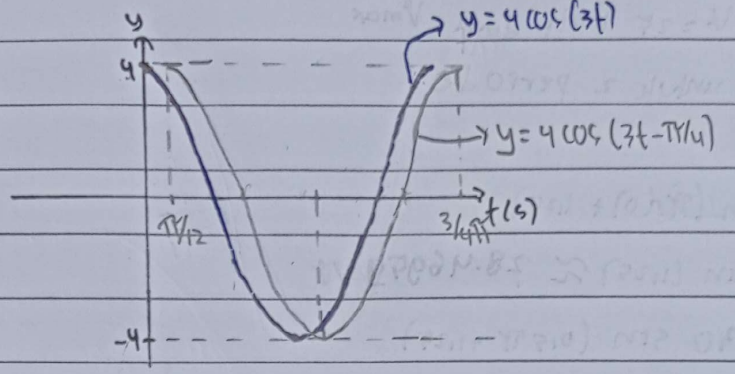
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3. $y = 4\cos(3t - \pi/4)$

- Amplitudo = 4 satuan
- kecepatan sudut (ω) = 3
- Periode (T) = $\frac{2\pi}{\omega} = \frac{2\pi}{3} = \frac{2}{3}\pi$ detik
- pergeseran kekanannan sejauh

$3t - \pi/4 = 0$

$t = \pi/12$ sampai $3/4\pi$



D. Diketahui $y = 40\sin(2x - 30^\circ) = 30\sin(2x - 0{,}5236)$

a. Tentukan amplitudo, waktu periodik, frekuensi dan sudut phase y (leading atau lagging $y_r = 30\sin 2x$ dan

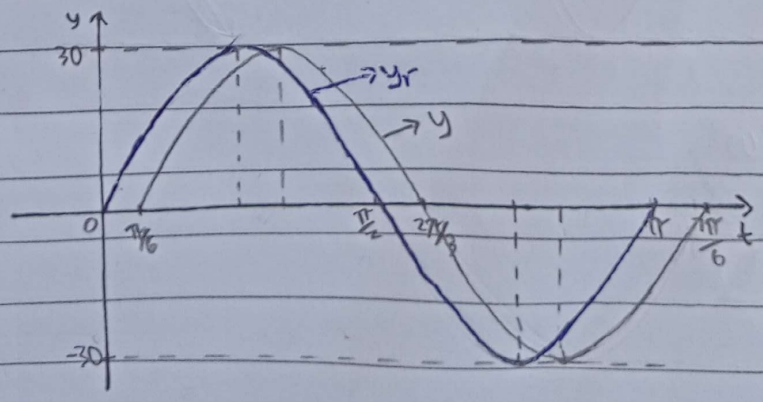
b gambarkan grafik tersebut.

- Jawab

a. $y = 30\sin(2x - 30^\circ)$ → (kayanya ada typo soalnya pak... saya ambil perkiraan itu karena 0,5236 radian ≈ 30°)

- Amplitudo = 30 satuan
- waktu periodik = $\frac{2\pi}{\omega} = \frac{2\pi}{2} = \pi$ detik
- Frekuensi = $f = \frac{1}{T} = \frac{1}{\pi}$ Hz
- sudut phase $y = a = 0{,}5236$ radian ≈ (30°)
- Grafik $y = 40\sin(2x - 30)$ terlambat / lagging sebesar 30° dibanding grafik $y_r = 30\sin 2x$

b. Grafik:



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2. Tegangan bolak balik suatu rangkaian listrik pada setiap saat t diberikan persamaan $V = 40 \sin(\pi t + 1{,}25)$ volt dengan t (second) > 0, Hitung

a) tegangan pada t = 0 dan t = 0,5 t

b) waktu (t) pada V = 25 volt dan Vmax

c) sketsa grafik V untuk 2 periode.

Jawab

a) $t = 0 \rightarrow V_{0t} = 40 \sin(\pi(0) + 1{,}25)$

$= 40 \sin(1{,}25) \approx 37{,}95938\ V$

$t = 0{,}5t \rightarrow V_{0{,}5t} = 40 \sin(0{,}5\pi + 1{,}25)$

$= 40 \sin(2{,}8208)$

$= 12{,}61289\ V$

b) $V = 25$ $\quad V_{max} = \text{Amplitudo} = 40V$

$25 = 40 \sin(\pi t + 1{,}25)$

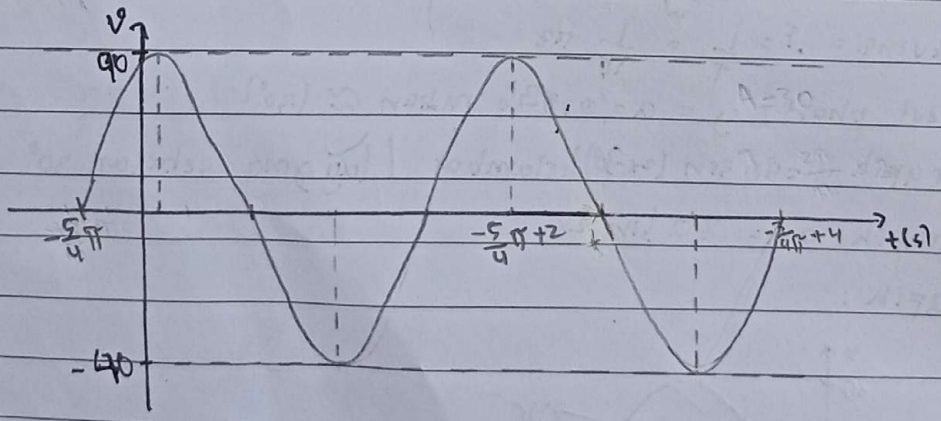
$\frac{25}{40} = \sin(\pi t + 1{,}25)$

$\sin^{-1}(5/8) = \pi t + 1{,}25$

$0{,}675130 - 1{,}25 = \pi t$

$t = \frac{0{,}67513 - 1{,}25}{\pi} \Rightarrow -0{,}182987\ t$

c) gambar grafik V untuk 2 periode $V = 40 \sin(\pi t + 1{,}25)$



A = 40

$\omega = \pi$ rad/s

$F = \frac{1}{\pi}$ rad

Pergeseran

$\pi t + 1{,}25 = 0$

$t = -\frac{5}{4\pi}$ kekiri

0,18254

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